

WorkForcePro QA Test Plan

Workforce Management & OJT Platform

Project: WorkForcePro

Version: 1.0

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Role: Sr. QA Engineer

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1. Purpose

This document defines the overall testing strategy, scope, approach, resources, deliverables, entry/exit criteria, risks, and schedule for validating the WorkForcePro Workforce Management and On-the-Job Training (OJT) platform prior to release.

2. Objectives

The objectives of this test plan are to:

- Validate core Employee Management functionality, including record creation, uniqueness rules, and department assignment.
- Validate end-to-end OJT workflows: training assignment, employee completion, and supervisor approval.
- Validate Certification Management, including multi-certification support and expiration tracking.
- Validate the platform's REST APIs for correctness, error handling, and adherence to business rules.
- Validate database integrity through direct SQL verification of employee, training, and certification data.
- Confirm the application is ready for automated regression coverage via Cypress.
- Confirm baseline accessibility of key user-facing screens.
- Provide a documented, traceable basis for the overall release-readiness decision.

3. Scope

3.1 In Scope

- Employee Management — creation, unique-email validation, department assignment.
- OJT Training — assignment, completion, supervisor approval/rejection workflow.
- Certification Management — multiple certifications per employee, expiry tracking, 30-day expiry alerts.
- API Testing — Employee and Training endpoints (positive, negative, and business-rule validation) via Postman.
- SQL Validation — direct database queries confirming employee, training, and certification data integrity.
- Cypress Automation — UI regression coverage for employee, training, and certification workflows.
- Accessibility Testing — keyboard navigation, form labeling, color contrast, and screen-reader basics on the reference login/product pages.

3.2 Out of Scope

- Performance and load testing.
- Penetration/security testing.
- Localization and internationalization testing.

4. Test Types

- Smoke Testing — verifies critical paths are functional after each build/deployment.
- Functional Testing — verifies each requirement behaves as specified, including negative and boundary cases.
- Regression Testing — re-verifies previously working functionality after changes, primarily via Cypress.

- API Testing — verifies request/response contracts, status codes, and business rules at the service layer.
- Database Testing — verifies data persistence, integrity, and correctness directly via SQL.
- UI Testing — verifies field-level behavior, validation messages, and workflow states in the browser.
- Cross-Browser Testing — verifies consistent behavior across Chrome and Edge (latest versions).
- Accessibility Testing — verifies keyboard, form, visual, and screen-reader basics against WCAG principles.
- Exploratory Testing — unscripted, experience-based testing to surface issues not covered by scripted cases.

5. Test Environment

Category	Details
Browsers	Chrome (latest), Edge (latest)
Operating System	Windows 11
API Testing Tool	Postman
Automation Framework	Cypress
Database	SQL database (per provided schema: employees, trainings, employee_trainings, certifications)
Version Control / Submission	GitHub
Test Data	Seeded QA dataset — sample employees, departments, supervisors, training modules, certifications — reset between cycles for reproducibility

6. Roles & Responsibilities

Added to clarify ownership across the testing effort referenced in the Purpose section ("resources").

Role	Responsibility
Sr. QA Engineer (Puskar Adhikari)	Test planning, test case design, manual execution, API/SQL/automation scripting, defect reporting, accessibility review
QA Lead	Reviews and approves the test plan, prioritizes defects, signs off on release readiness
Developer(s)	Fixes defects, supports root-cause investigation, reviews API/DB contracts
Product Owner	Clarifies requirements, confirms expected behavior for ambiguous rules (e.g., mandatory fields)

7. Entry Criteria

- A stable build of WorkForcePro is deployed to the QA environment.
- Requirements (EMP-001–003, OJT-001–004, CERT-001–003) are documented and available.
- Test data is prepared — employees, departments, supervisors, training modules.
- APIs are accessible and reachable from Postman.

8. Exit Criteria

- All critical defects are resolved and verified.
- All smoke tests pass on the final build.
- No open blocker defects remain.

- Agreed test coverage (100% of documented requirements mapped and executed per the RTM) is completed.

9. Test Schedule / Milestones

Added to support the schedule commitment referenced in the Purpose section, mapped to the assignment's estimated 4–6 hour effort.

Phase	Activity	Target
1	Test planning, scenario/test case design, RTM	Day 1
2	Manual functional testing & defect logging	Day 1–2
3	API testing (Postman collection + validation scripts)	Day 2
4	SQL validation queries	Day 2
5	Cypress automation suite	Day 2–3
6	Accessibility testing (SauceDemo)	Day 3
7	Final review, test summary report, submission	Day 3

10. Deliverables

- Test Plan (this document)
- Test Scenarios
- Test Cases
- Requirement Traceability Matrix (RTM)
- Smoke Test Checklist
- Defect Report
- Postman Collection
- SQL Queries
- Cypress Automation Suite
- Accessibility Report
- Test Summary Report

11. Risks

- Incomplete or ambiguous requirements (e.g., whether Department is mandatory at creation).
- Unstable test environment causing inconsistent or non-reproducible results.
- Unavailable or insufficient test data (departments, training modules, certifications).
- API changes introduced mid-cycle without corresponding documentation updates.
- Schedule constraints given the assignment's estimated 4–6 hour testing window versus full coverage depth.

12. Assumptions

- Employee emails are enforced as unique by the system (EMP-002).
- Supervisor approval is mandatory before a completed training is considered finalized (OJT-003).
- Certifications are tracked by an expiry date, and the system evaluates a 30-day expiry window against the current server date (CERT-002, CERT-003).

- Only users with a Supervisor role can assign training and approve/reject completions (OJT-001, OJT-003).

13. Approval

Role	Name	Signature / Date
Sr. QA Engineer	Puskar Adhikari	2026-07-05
QA Lead		

14. High-Level Test Scenarios

Summarized per requirement; forms the basis for the detailed test cases and RTM below.

Req. ID	Scenario	Description
EMP-001	Employee record creation	Verify employees can be created with valid mandatory data, and that creation fails gracefully when mandatory fields are missing or invalid.
EMP-002	Unique email enforcement	Verify the system prevents two employee records from sharing the same email address, and allows unique emails without restriction.
EMP-003	Department assignment	Verify employees can be assigned to a department at creation and reassigned afterward, and verify system behavior when no department is selected.
OJT-001	Assign OJT training	Verify a supervisor can assign a training module to an employee, and that assignment is restricted to authorized roles (supervisors).
OJT-002	Mark training as completed	Verify an employee can mark an assigned training as completed with required completion data, and cannot do so with incomplete data.
OJT-003	Supervisor approval gating	Verify completed training remains in a pending state until a supervisor explicitly approves or rejects it.
OJT-004	Unassigned training restriction	Verify an employee cannot mark as complete a training module that was never assigned to them.
CERT-001	Multiple certifications per employee	Verify a single employee record can hold and display more than one certification simultaneously.
CERT-002	Certification expiry tracking	Verify every certification record stores and displays an expiration date, and that the field is mandatory.
CERT-003	Expiring-certification detection (30-day window)	Verify the system correctly flags certifications expiring within 30 days, including boundary conditions at exactly 30 and 31 days.

15. Detailed Test Cases

Each test case is traced to a specific requirement ID. Equivalent API-level variants of the negative/boundary cases are exercised again in the Postman collection deliverable.

TC ID	Req ID	Preconditions & Steps	Expected Result	Priority
TC-001	EMP-001	Create employee with valid data Precondition: User is logged in with permission to create employees. 1. Navigate to Employee Management > Add Employee. 2. Enter valid Name, Employee ID, Email, Department, Supervisor. 3. Click Save.	Employee record is created and appears in the employee list with all entered details.	High
TC-002	EMP-001	Create employee with missing Name field Precondition: User is on the Add Employee form. 1. Leave Name blank; fill all other fields validly. 2. Click Save.	System blocks submission and displays a validation error indicating Name is required. No record is created.	High
TC-003	EMP-001	Create employee with missing Email field Precondition: User is on the Add Employee form. 1. Leave Email blank; fill all other fields validly. 2. Click Save.	System blocks submission and displays a validation error indicating Email is required. No record is created.	High
TC-004	EMP-002	Create employee with duplicate email Precondition: An employee already exists with email 'jane.doe@company.com'. 1. Create a new employee using the same email. 2. Click Save.	System rejects the submission with a clear 'email already exists' error. No duplicate record is created.	High
TC-005	EMP-002	Create employee with same name, different email Precondition: An employee 'John Smith' already exists. 1. Create another employee named 'John Smith' with a unique email. 2. Click Save.	Record is created successfully — name duplication is allowed, only email uniqueness is enforced.	Medium
TC-006	EMP-003	Assign department during employee creation Precondition: At least one department exists in the system. 1. Create an employee and select a Department from the dropdown. 2. Click Save.	Employee is created and correctly associated with the selected department.	High
TC-007	EMP-003	Attempt to create employee without selecting a department Precondition: User is on the Add Employee form. 1. Fill all fields except Department. 2. Click Save.	System either blocks submission with a validation message (if department is mandatory) or creates the record as 'Unassigned' — behavior should be explicit; flag as defect if undefined/inconsistent.	Medium
TC-008	EMP-003	Reassign employee to a different department Precondition: Employee exists in Department A. 1. Edit the employee record. 2. Change Department to Department B and Save.	Employee's department is updated to Department B; historical reports reflect the change going forward.	Medium
TC-009	OJT-001	Supervisor assigns training module to employee Precondition: Logged in as a Supervisor; employee and training module exist. 1. Navigate to Training Assignment. 2. Select Employee, Training Module, and Due Date. 3. Click Assign.	Training assignment is created and visible on the employee's training record with status 'Assigned'.	High
TC-010	OJT-001	Non-supervisor attempts to assign training Precondition: Logged in as a standard Employee (non-supervisor). 1. Attempt to access Training Assignment. 2. Attempt to assign a training module.	Action is blocked; system denies access or hides the assignment control, confirming role-based restriction.	High
TC-011	OJT-001	Assign the same training module to an employee twice Precondition: Employee already has Training Module X assigned and not yet completed. 1. Attempt to assign Training Module X again to the same employee.	System either prevents duplicate active assignment with a warning, or clearly supports multiple concurrent assignments by design — flag as defect if it silently creates confusing duplicates.	Medium
TC-012	OJT-002	Employee marks assigned training as completed Precondition: Employee has Training Module X in 'Assigned' status. 1. Open the assigned training. 2. Enter Completion Date and Notes. 3. Click Mark as Completed.	Training status changes to 'Pending Approval'; Completion Date and Notes are saved.	High
TC-013	OJT-002	Mark training completed without a Completion Date Precondition: Employee has an assigned training open for completion. 1. Leave Completion Date blank. 2. Click Mark as Completed.	System blocks the action and displays a validation message requiring Completion Date.	Medium

TC ID	Req ID	Preconditions & Steps	Expected Result	Priority
TC-014	OJT-003	Supervisor approves completed training Precondition: Training is in 'Pending Approval' status; user logged in as the assigned Supervisor. 1. Open the pending training record. 2. Click Approve.	Training status changes to 'Approved'/'Completed'; record reflects supervisor and approval timestamp.	High
TC-015	OJT-003	Supervisor rejects completed training Precondition: Training is in 'Pending Approval' status. 1. Open the pending training record. 2. Click Reject, optionally add a reason.	Training status changes to 'Rejected'; employee is notified and can resubmit or is routed per business rule.	High
TC-016	OJT-003	Verify training cannot be marked Completed/Approved without supervisor action Precondition: Employee has marked training as completed. 1. Without supervisor action, check the training status via UI and/or API.	Status remains 'Pending Approval' — it never auto-transitions to 'Approved' without an explicit supervisor action.	Medium
TC-017	OJT-004	Employee attempts to complete a training that was never assigned Precondition: Employee has no assignment for Training Module Y. 1. Attempt to mark Training Module Y as completed (via UI navigation or direct API call).	System rejects the action with an error indicating the training is not assigned to this employee. No completion record is created.	High
TC-018	CERT-001	Employee holds multiple certifications Precondition: Employee record exists. 1. Add Certification A with a valid expiry date. 2. Add Certification B with a different valid expiry date.	Both certifications are saved and displayed independently on the employee's certification list.	Medium
TC-019	CERT-002	Add certification with a valid expiration date Precondition: Employee record exists. 1. Navigate to Certifications > Add. 2. Enter certification name and a future expiry date. 3. Save.	Certification is saved with the expiry date correctly stored and displayed.	High
TC-020	CERT-002	Add certification without an expiration date Precondition: User is on the Add Certification form. 1. Leave Expiry Date blank. 2. Save.	System blocks submission with a validation error, since expiry date must be tracked per CERT-002.	Medium
TC-021	CERT-003	System flags certification expiring within 30 days Precondition: Certification exists with expiry date 15 days from today. 1. Open the Certification/Compliance dashboard.	Certification appears in the 'Expiring Soon' list/report, clearly flagged.	High
TC-022	CERT-003	System does not flag certification expiring in 45 days Precondition: Certification exists with expiry date 45 days from today. 1. Open the Certification/Compliance dashboard.	Certification does NOT appear in the 'Expiring Soon' list, confirming the 30-day window is correctly bounded.	Medium
TC-023	CERT-003	Boundary test — certification expiring exactly 30 days from today Precondition: Certification exists with expiry date exactly 30 days from today. 1. Open the Certification/Compliance dashboard.	Certification appears in the 'Expiring Soon' list (inclusive boundary) — confirm and document actual system behavior as this is a common off-by-one defect source.	Medium

16. Requirement Traceability Matrix (RTM)

Confirms 100% requirement coverage — every documented requirement maps to at least one executed test case, satisfying the Exit Criteria in Section 8.

Requirement ID	Requirement Description	Mapped Test Cases
EMP-001	Users can create employee records.	TC-001, TC-002, TC-003
EMP-002	Employee email addresses must be unique.	TC-004, TC-005
EMP-003	Employees can be assigned to a department.	TC-006, TC-007, TC-008
OJT-001	Supervisors can assign OJT training modules to employees.	TC-009, TC-010, TC-011
OJT-002	Employees can mark assigned training as completed.	TC-012, TC-013
OJT-003	Completed training requires supervisor approval.	TC-014, TC-015, TC-016
OJT-004	Employees cannot complete training that has not been assigned.	TC-017
CERT-001	Employees can hold multiple certifications.	TC-018
CERT-002	Certification expiration dates must be tracked.	TC-019, TC-020
CERT-003	System identifies certifications expiring within 30 days.	TC-021, TC-022, TC-023

Coverage summary: 10 / 10 requirements mapped (100%). 23 test cases total — 10 positive, 8 negative, 5 boundary/edge.